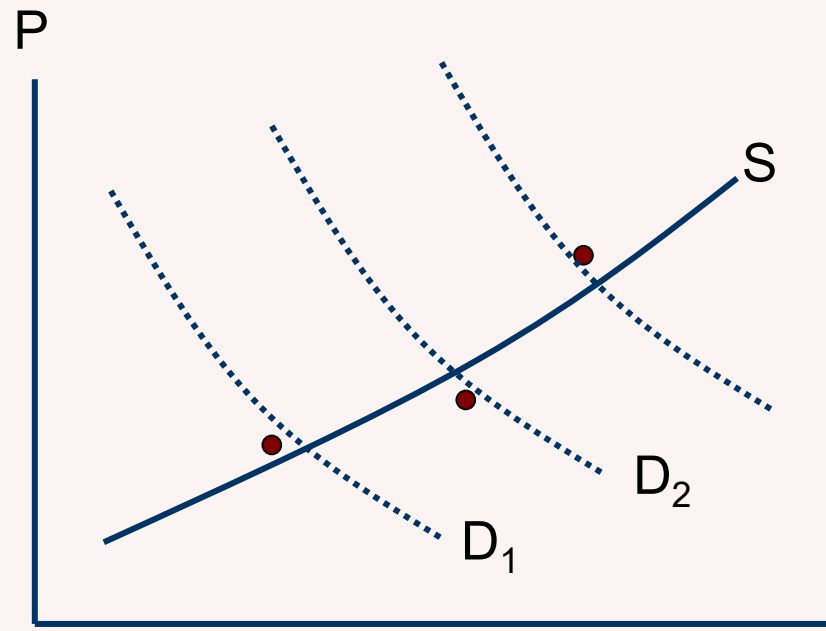


Topic 3—**Seminar 3 Addon** (Week 5):  
Identifying the Housing Supply Price Elasticity for  
UK Regions: **The Bartik Instrument**



# The Bartik Instrument

- Proposed by Timothy Bartik in his 1991 book
  - ▶ “Who Benefits from State and Local Economic Development Policies? W.E. Upjohn Institute.”
- Also called
  - ▶ Shift-share approach
  - ▶ Labor demand shock measure
- Definition
  - ▶ Inner product of the (past) industry-location employment **shares** (the ‘share’) and the (national-level) industry component of the (employment) growth rates (‘the shift’)
  - ▶ The *predicted* local employment

# Formally

- $\hat{B}_l = \sum_k z_{lk} g_k$
- Where:
  - ▶ Location  $l=1, \dots, L$
  - ▶ Industry  $k=1, \dots, K$
  - ▶  $z_{lk}$ : Share of industry  $k$  in location  $l$  (*must pre-date sample period*)
  - ▶  $g_k$ : National-level employment growth rate in industry  $k$

# A relevant example: Hilber & Vermeulen (2016)

- Consider 353 local jurisdictions ( $L=353$ ) and 7 industries ( $K=7$ )
- Industry shares are from 1971 Census (**pre-dating sample period**)
- A national time series of employment growth by industry from 1971 to 2008 was obtained by combining Census of Employment from 1971-1978 and ONS data from 1978-2008
- Use this data to create predicted employment from 1974 to 2008 (**sample period**)
- Formally:  $LLD_{lt} = \sum_{k=1}^7 emp_{l,1971}^k \times index_t^k$ ,
  - ▶ Where  $emp_{l,1971}^k$  is employment in location  $l$  industry  $k$  in 1971, and  $index_t^k$  is the national-level employment index for industry  $k$  (base year 1971)
  - ▶ (Not exactly Bartik measure; but correlates perfectly with Bartik measure, so can be used interchangeably)

# Why? (In the case of Hilber & Vermeulen 2016)

- Local earnings are not ideal demand shifter
    - ▶ Correlated with construction salaries (which shift S-curve)
    - ▶ Can be influenced by house prices through income sorting
      - Say high-income HHs sort into locations with tightening local regulation because they value amenities created by tightening regulation (e.g., nice views, open space, preserved cityscape)
      - If so, then changes in earnings may not only reflect demand shift but potentially also a supply shift
  - Bartik-instrument only uses local industry share information pre-dating sample period & national-level growth rates
    - ▶ Both are orthogonal to local income sorting over time & changes in construction wages
  - Could in principle use Bartik-measure as instrument for earnings in TSLS, but leads to too weak identification (KP F-stat too small), hence H&V (2016) use directly (in reduced form)
- ⇒ Bartik instrument preferable as demand shifter!

# References

- Proposed in
  - ▶ Bartik, T. (1991) *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute.
- Further reading
  - ▶ Goldsmith-Pinkham, P., I. Sorkin & H. Swift (2020) Bartik Instruments: What, When, Why and How. *American Economic Review*, Vol. 110. No. 8, 2586-2624.
  - ▶ Hilber, C. & W. Vermeulen (2016) The Economic Impact of Supply Constraints on House Prices in England. *Economic Journal*, Vol. 126, No. 591, 358-405. (See pages 374, 381-383 & 398-399)